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5.1 Práctica: Diseño de un controlador para un sistema de segundo orden

Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R i_1(s) + L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)]$$

$$L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)] = R i_2(s) + R i_2(s) + \frac{i_2(s)}{Cs}$$

$$V_s(s) = R i_2(s) + \frac{i_2(s)}{Cs}$$

Procedimiento algebraico

$$V_e(s) = (Ls + 2R) i_1(s) - (Ls + R) i_2(s)$$

$$i_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R \right) i_2(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} i_2(s)$$

$$V_s(s) = \left(R + \frac{1}{Cs} \right) i_2(s) = \frac{CRs + 1}{Cs} i_2(s)$$

$$V_e(s) = \left[(Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} - \frac{(Ls + R)}{Cs} \right] i_2(s)$$

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{(CRs+1) I_2(s)}{(Ls+2R) I_1(s) - (Ls+R) I_2(s)}$$

$$\therefore I_2(s) = \frac{CLs^2 + 3(CR+1) I_1(s)}{(Ls^2 + CRs)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{-CRs+1}{Ls} I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{3CLR^2 + (5CR^2+L)s + 2R}{Ls^2 + CRs} I_2(s)$$

$$= \frac{(CR+1)(Ls+R)}{3CLR^2 + (5CR^2+L)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(LRs^2 + (CR^2+L)s + R)}{3CLR^2 + (5CR^2+L)s + 2R}$$

$$CR = 0.022748$$

$$R = 2200$$

$$R = 2.2K\Omega$$

$$CR^2 + L = 2274.822$$

$$C = 470\mu F$$

$$L = 22mH$$

$$\frac{V_s(s)}{V_e(s)} = \frac{0.022748s^2 + 2274.822s + 2200}{150.1368s^2 + 11374.022s + 4400}$$

$$3(CR)^2 = 150.1368$$

$$5(CR^2+L) = 11374.022$$

$$2R = 4400$$

Error estacionario a un escalon unitario

$$H(s) = \left(\frac{H_s(s)}{H_c(s)} \right) \frac{1}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \left[1 - \frac{0.022748s^2 + 2274.822s + 2200}{150.1368s^2 + 11374.022s + 4400} \frac{1}{s} \right]$$

* Se cancela s

$$e_{ss} = 1 - \frac{2200}{4400} = 1 - 0.5 = 0.5 \text{ o } 50\%$$

Estabilidad en lazo abierto

$$R = 2.2 \text{ k}\Omega$$

$$L = 22 \text{ mH}$$

$$C = 470 \text{ nF}$$

Calcular raíces del denominador o polos del sistema

$$3CLR^2 + (sCR^2 + L)s + 2R = 0$$

$$\lambda_1 = -0.386$$

$$a = 3(2.2 \times 10^3)(470 \times 10^{-6})(22 \times 10^{-6})$$

$$\lambda_2 = -166.66 \times 10^6$$

$$b = [s(2.2 \times 10^3)^2(470 \times 10^{-6})] + 22 \times 10^6$$

$$c = 2(2.2 \times 10^3)$$

El sistema es estable con una respuesta sobreamortiguada.

Bosquejo de la respuesta (transitorio + estado estacionario)

